

Emily Fairfield

Data Scientist

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PROFESSIONAL AWARDS

2024 General James Ferguson Engineering Award Finalist | 2023 Air Force Civilian Achievement Award | 2022 Junior Engineer of the Year

SKILLS

Machine Learning | Data Visualization (D3, Matplotlib) | Extract, Transform, Load (ETL) | Project Management | Unit Testing | Strategic Planning | Process Improvement | Technical Requirements Definition | Code Documentation | Python (SciKit Learn, Dask, Pandas, Numpy, Matplotlib) | MLOps (MLflow, DVC, DagsHub, Git) | SQL | VBA | Process Flow Mapping | Natural Language Processing

WORK EXPERIENCE

US AIR FORCE CIVILIAN SERVICE 2017 - Present

DATA SCIENTIST, Special Operations Forces (SOF) US AFLCMC | Robins Air Force Base (AFB), GA | 2023 - Present

- **Multithreaded, Vectorized MIL-STD 1553B Bus Data Translator** - Developed, tested, and documented multithreaded, vectorized data translator that converts raw byte streams into human-readable data for real-time aircraft health monitoring and intrusion detection on Special Operations aircraft. (Petabytes of data annually). Compared to legacy translator, simultaneously improved runtime by 10x, quantity of data translated by 6x, and output file size by 13x, enabling real-time translation on a typical laptop.
- **Term Frequency-Inverse Document Frequency (TF-IDF) and K-Means Clustering** - Applied TF-IDF & K-Means Clustering to maintenance records to find prominent issues by aircraft delivery date for contractor quality standards accountability.
- **Statistical Analyses** - Applied Kolmogorov-Smirnov test to find time ranges of steady level flight for fuel efficiency project.
- Assess & validate contractor-developed AI applications & train engineers of other disciplines to generate AI requirements.
- Work with engineers & users to confirm understanding of data & ensure model outputs are actionable. Train junior data scientists.

DATA SCIENTIST, Automatic Test Systems (ATS) US AFLCMC | Robins AFB, GA | 2019 - 2023

- Led technical innovation efforts for internal & external customers. Identified opportunities, executed projects, & trained users.
- **XML Data Translator** - Developed script to clean & transform avionics troubleshooting test result files for customers' ML models.
- **Nuclear Certification Database** - Developed database with MS Access, VBA, & SQL. Incorporated feedback. Wrote manual.
- **Cameo Systems Models** - Adapted models for ATS & trained hundreds of users. Created promotional video.

PROCESS ENGINEER, F-15 Maintenance US Air Force Sustainment Center | Robins AFB, GA | 2017 - 2019

- **Capacity Improvement Project** - Eliminated 3 tows/F-15 from the maintenance process. (+66 days of F-15 Aircraft Availability (AA)/year). Increased F-15 inspection & repair capacity by 150% without building a new facility, accommodating new F-15 modification programs. (+600 days of F-15 AA/yr / +\$30M/yr workload). Regularly led process flow mappings with stakeholders.

MANAGEMENT ENGINEER Northside Hospital | Atlanta, GA | 2016 - 2017

- **Labor & Delivery (L&D) Room Turnover** - Reduced 70% of L&D patients' stay by 21 min, with cascading benefit to waiting room.
- Managed productivity models for >100 departments, including surgical, postpartum, radiology, and behavioral health.
- Served as consultant to hospital managers, training them to use productivity software & collaborating on process improvements.

MACHINE LEARNING & OPTIMIZATION PROJECTS

Personal Projects - Sample of ML (PyTorch), Data Cleansing/ETL, & Optimization Projects: <https://emilyfairfield.github.io/> 2025

SMART on FHIR App - Captured health data & exported to FHIR-compliant format for interoperability with simulated EHR system. 2022

Machine Learning Master's Program Projects - Georgia Institute of Technology 2022-2023

Developed dozens of ML algorithms (Supervised, Unsupervised, Reinforcement Learning) including data cleansing, feature engineering, accuracy metric selection, & hyperparameter tuning. Conducted experiments to identify overfitting/underfitting & accuracy by class.

Logistics Decision Support System - Georgia Tech Senior Design (Consultants for Gypsum Management & Supply) 2015

Created a Python-based logistics tool that plans truck loading & routing while allowing last-minute orders. Delivery cost savings of 6.5%.

EDUCATION

MASTER OF SCIENCE, COMPUTER SCIENCE, ML CONCENTRATION | Georgia Institute of Technology | GPA: 3.8/4.0 Dec. 2023

MASTER OF BUSINESS ADMINISTRATION (MBA) | Kennesaw State University | GPA: 4.0/4.0 May 2020

BACHELOR OF SCIENCE, INDUSTRIAL & SYSTEMS ENGINEERING | Georgia Institute of Technology | GPA 3.38/4.0 Dec. 2015